

1. Stay where you are, list all the files and folders in your current directory (including hidden)

```
yash@Yashwanth:~$ ls -a
.          .cache          .gitconfig      .node_repl_history .sudo_as_admin_successful CFI-A26-Yash
..         .calc_history   .landscape      .npm              .viminfo           learning-git
.bash_history .config        .lessht       .nvm              .vscode-server     portfolio
.bash_logout .copilot       .local         .profile          .wakatime          practice-git
.bashrc      .dotnet        .motd_shown    .ssh              .wakatime.cfg      prephase1
yash@Yashwanth:~$
```

2. In one command, create dir1/dir2/dir3. (dir3 is a subdirectory from dir2, and dir2 is a subdirectory from dir1).

```
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux$ mkdir -p dir1/dir2/dir3
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux$ tree
.
├── dir1
│   └── dir2
│       └── dir3
```

3. Linux command to download a file from a URL? Ex:

```
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux$ curl -O https://www.w3schools.com/html/img_chania.jpg
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 45427 100 45427 0 0 616k 0 --:--:-- --:--:-- --:--:-- 624k
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux$ ls
dir1 img_chania.jpg
```

https://www.w3schools.com/html/img_chania.jpg

4. What is the command to go to the parent directory of the current directory?

```
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux$ cd ..
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1$
```

5. What is an inode number

Inode stores the metadata for the data, inode full form is index node it can be accessed as a flag -i

6. What is the one command to remove any directory and all files in it?

```
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2/dir3$ touch file.txt file1.txt
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2/dir3$ ls
file.txt file1.txt
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2/dir3$ cd ..
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2$ rm -rf dir3/
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2$ ls
```

7. Write a Script to Display the last line of a text file, ex: from the file /etc/passwd.

```
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2$ tail -1 file1.txt  
line 5
```

8. Use the date command to display the date in YYYY/MM/DD format.

```
yash@Yashwanth:~/CFI-A26-Yash/Phase-Assignments/Phase-1/linux/dir1/dir2$ date +%Y/%m/%d  
2026/03/01
```

9. Explain the difference between:

- cp and mv

Cp is used to copy a file it makes a duplicate for the original file and it is stored in the destination file

Mv is used to move a file from one source to a destination and it also allows to rename the files

- rm and rmdir

Rm is used to remove files and directories when empty unless flags are used

Rmdir can only be used on empty directories and cannot delete when any files are in it

10. Explain the difference between: a) apt update and apt upgrade

- Apt update is used to update the current version of the package
- Apt upgrade checks for any new versions available for the package to install them

11. What is the difference between absolute path and relative path

- Absolute path is the path which can be accessed from predefined paths like ~ home or / that already exist in the system
- Relative path is the path from the pwd to the destination

12. What is the command to install package in Ubuntu/linux command line

- sudo apt install <package_name>